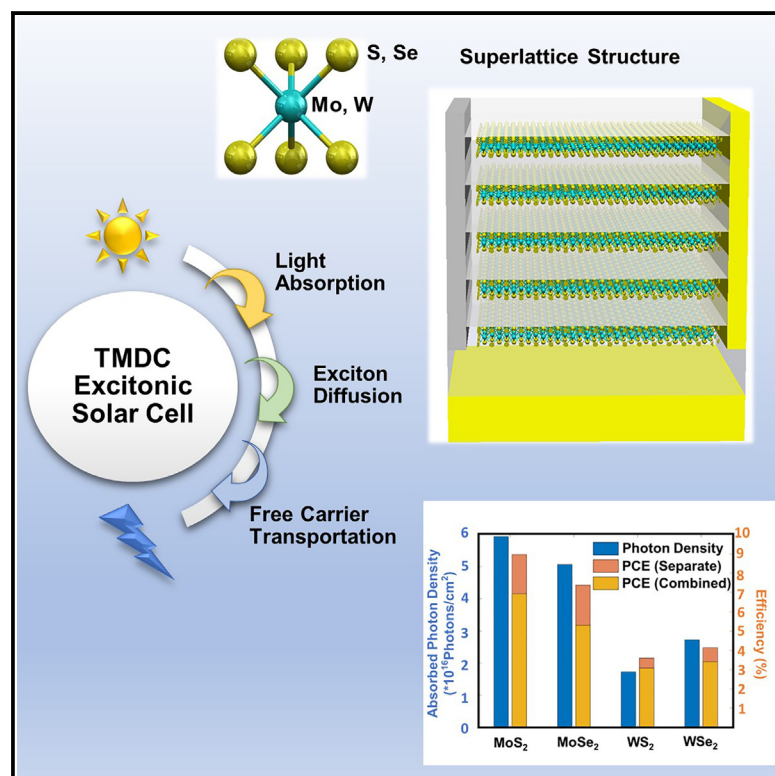


How good can 2D excitonic solar cells be?

Graphical abstract



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In brief

This study shows a comprehensive design and modeling of monolayer 2D transition metal dichalcogenide-based photovoltaic devices. Electronic, photonic, and excitonic properties of the semiconductors have been accounted for and optimized to predict the maximum theoretical performance and device design parameters. A 12.87% power conversion efficiency and a specific power of over 100 W/g were achieved for the optimized device configuration.

Highlights

- Comprehensive theoretical design and analysis of 2D excitonic photovoltaics (PVs)
- Concurrent optimization of both optical and electronic properties
- Optimized PVs from MoS₂ achieve 12.87% power conversion efficiency
- MoS₂ superlattice absorber-based PV devices can achieve a specific power >100 W/g



Understand

Early stage research on device properties, design, and physics

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Article

How good can 2D excitonic solar cells be?

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THE BIGGER PICTURE This work explores the potential of two-dimensional transition metal dichalcogenides (TMDCs) as excitonic semiconductors for photovoltaic applications. While TMDCs have been heavily investigated in photodetector and photovoltaic devices, their ultimate efficiency limits are unknown, and the impact of excitons have been unaccounted for in studies thus far. This article provides a comprehensive theoretical design of TMDC-based excitonic PV devices, identifying limiting factors and optimal geometric design parameters accounting for electronic, photonic, and excitonic effects. The proposed designs hold promises for achieving a solar to electrical power conversion efficiency >12% concurrently with specific power of over 100 W/g. Such performance opens potential applications in aerospace, remote sensing, and wearable electronics.

SUMMARY

Excitonic semiconductors have been a subject of research for photovoltaic (PV) applications for many decades. While organic photovoltaics have now exceeded 19% power conversion efficiency (PCE), the advent of inorganic semiconductors such as two-dimensional transition metal dichalcogenides (TMDCs) has renewed interest in excitonic solar cells due to their sub-nm film thicknesses. While several reports have been published on TMDC-based photovoltaics, achieving PCEs higher than 6% has remained challenging. In this study, we perform a comprehensive analysis of the optical and electronic characteristics of TMDC-based excitonic PV devices. Our designs are consciously chosen with the consideration of manufacturing challenges associated with large-area photovoltaic technologies. Our analysis suggests that whereas the PCE for 2D excitonic solar cells may be limited to <13%, a specific power >100 W g⁻¹ may be achieved with our proposed designs, making them attractive in aerospace applications, distributed remote sensing, and wearable electronics.

INTRODUCTION

Thin-film photovoltaics (PVs), such as those based on III-V semiconductors, CdTe, and 3D perovskites, have been a source of sustained research and commercial interest. However, they occupy a small share of the large-scale, grid-tied market due in part to lack of production scaling. Hence, electricity prices remain high compared with the current dominant photovoltaic technology of silicon photovoltaics (PVs).¹ Silicon is also the second most abundant element in the Earth's crust, making it easier for manufacturers to access raw materials than III-V materials and CdTe. This feature is a critical determinant in the scalability of production, rendering silicon wafers the most desirable option for mass production. However, thin-film PVs have long been considered a potential solution in applications that require lightweight solar cells, such as aerospace applications, remote sensors, and wearable electronics.^{2–4} These applications require lightweight, high-power-density solar cells that have moderate scalability. While new

and emerging materials such as organic,⁵ chalcogenides,^{6–9} and 2D organic-inorganic perovskites¹⁰ are also being heavily investigated, they typically use active layers that are on the order of microns; this is an improvement over silicon in terms of both power density and weight, but does not represent a fundamental limit of weight. Instead, 2D excitonic semiconductors present the possibility of solar cells with active layer thicknesses of a few atoms. In particular, 2D transition metal dichalcogenides (TMDCs) of MX₂ (M = Mo, W and X = S, Se, Te) have recently gained traction for lightweight PV applications. Their large optical constants result in large loss-tangent values across the visible region; this, when combined with their availability in high optical and electronic quality over wafer scales, makes them increasingly viable candidates for thin-film, ultralight-weight PVs.^{11,12}

Further, a notable feature of 2D TMDC semiconductors is a transition from an indirect band gap in bulk to the direct band gap in monolayers. This enables a high photoluminescence quantum yield and thus a high radiative efficiency.¹³ Finally,



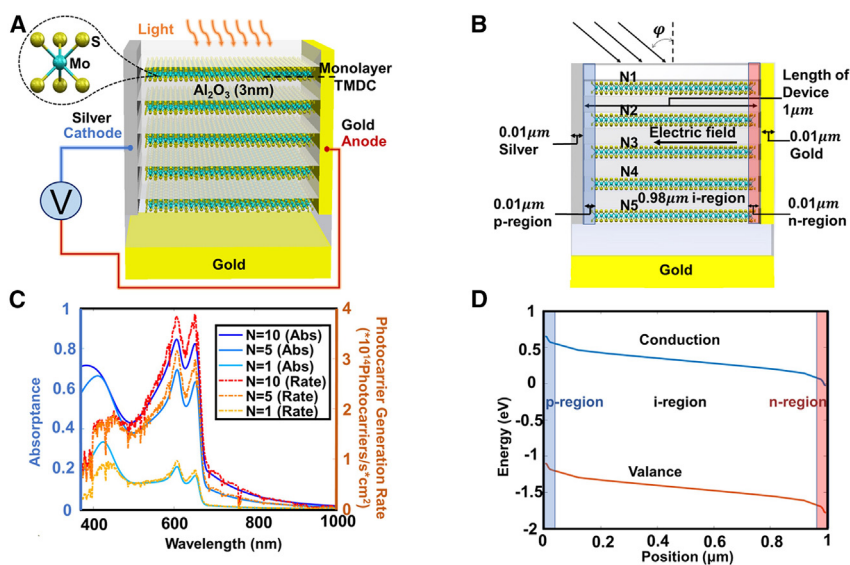


Figure 1. The superlattice structure enables high photon absorption

(A) Schematic model of the superlattice structure. (B) Side view of the model with labeled p-i-n regions and layers. (C) Simulated absorbance spectra and photocarrier generation rate for MoS₂/Al₂O₃ (N = 1, 5, 10). (D) Energy band diagram of the monolayer MoS₂ for p-i-n junction.

the wide range of band gaps (1–2.5 eV¹⁴) across TMDC materials and their favorable van der Waals bonding for facile heterointegration make TMDCs attractive candidates for single-junction, tandem, and multi-junction solar cells.¹⁵

As a result of these ideal characteristics, several studies reporting microscale PV devices from bulk and monolayer TMDCs are available. However, the power conversion efficiency (PCE) values of these experimentally reported TMDC solar cells are typically lower than 2%,^{16–18} whereas the highest PCEs reported are 9.03% in MoS₂¹⁹ and 6.3% in WS₂.²⁰ Per the detailed balance model,¹⁵ thin-film, single-junction TMDC solar cells can reach up to 27% PCE, which is comparable to crystalline silicon. This vast disparity between theoretical maximum and experimentally observed values therefore merits the question: how good can TMDC solar cells be made in terms of PCE and power density? To answer this question, we developed a combined optical and electronic model to simulate the PV characteristics of four TMDC materials (MoS₂, WS₂, MoSe₂, and WSe₂) based on an experimentally reported, large-area, scalable superlattice structure.²¹ Using our model, we optimized the parameters and performance of a monolayer MoS₂-based PV device with combined contacts, attaining a PCE as high as 9.22% under one-sun (AM1.5) illumination with a specific power exceeding 100 W/g. Subsequently, we performed a benchmarking analysis of our 2D PV devices against both TMDC-based PVs and other leading technologies reported in literature with respect to the performance metrics of PCE and specific power. Our findings indicate that 2D TMDC-based PVs, when optimized for both optical and electronic design, exhibit superior performance compared with other thin-film materials in terms of power density, which is vital for many applications such as aerospace, remote sensing, and wearable technology.

RESULTS AND DISCUSSION

Device structure and optimization of photon absorption

The proposed 2D TMDC-based PV superlattice device is shown in Figure 1A. The device consists of a repeating unit cell of a

monolayer MoS₂ absorber (0.65 nm) and an Al₂O₃ insulator (3 nm) and is placed on top of an Al₂O₃/Au substrate with the Au serving as a reflector. The thickness of the Al₂O₃ layer has been optimized to enhance photocarrier generation. The active layer of the device is intrinsic (no doping) and is 0.98 μm long, with silver and gold cathode and anode electrodes measuring 0.01 μm in length each. The

heavily doped P region (10¹⁹ cm⁻³) near the cathode and the heavily doped n-region (10¹⁹ cm⁻³) near the anode are also 0.01 μm in length. The incident angle of the light is zero, meaning it is normal to the surface. Additional details regarding the device structure and parameters used for simulation are provided in the [methods](#) and in [Table S1](#).

By using the transfer matrix method (TMM) to calculate the absorbed photon density of each layer,²² we find that increasing the number of layers (N) increases the absorbed photon density in the 400–700 nm wavelength range as seen in Figure 1C. This increase is particularly significant when going from N = 1 to N = 5. Further increase from N = 5 to N = 10 does not lead to as dramatic an increase in broadband absorbance and actually decreases the absorbed energy per unit weight. Therefore, the electrical simulations for our proposed device were based on an N = 5 superlattice design. The energy band diagram for the active layer, which has a 1.80 eV band gap for monolayer MoS₂,²³ is shown in Figure 1D. The band diagram is that of a p-i-n lateral homojunction, taking into account the built-in field created by degenerate p-doping and n-doping near the contacts, which promotes charge carrier selectivity and improves device performance.²⁴

Role of excitons and their dissociation efficiency

Excitons dominate the optical response of semiconducting 2D TMDCs, not only in the monolayer limit but also in the bulk.²⁵ However, thus far, all attempts in literature to quantify and estimate theoretical PV potential for TMDCs have failed to treat excitons seriously in their models.^{26–28} Not only do excitons dominate the optical properties of low-dimensional semiconductors, but they have also been confirmed through both theory and experiment to limit the performance of other well-known excitonic semiconductor-based PVs, namely organic PVs (OPVs).^{25,29–33} Therefore, any attempt to seriously quantify the performance limits of semiconducting 2D TMDC-based PVs must include a detailed treatment of excitons. In this work, we have thus examined the effects of exciton binding energy

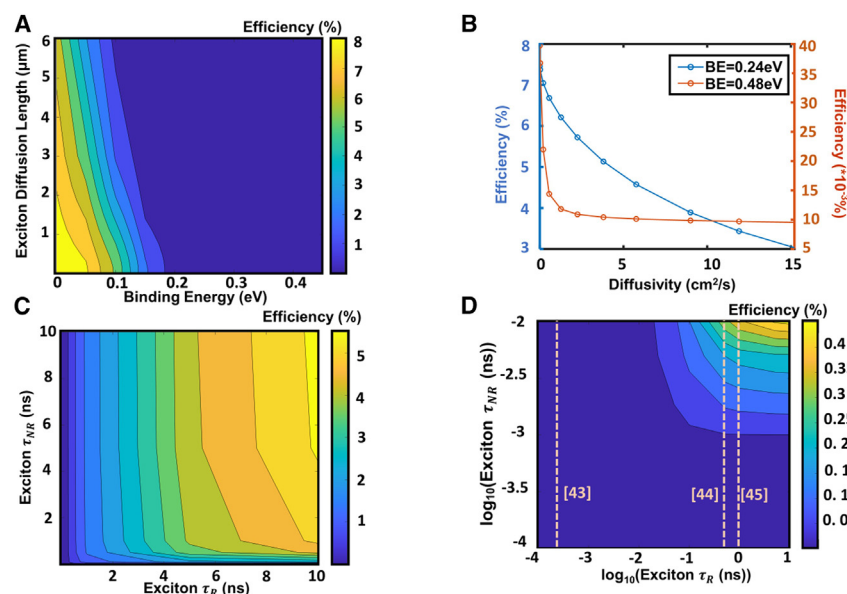


Figure 2. Optimization of excitonic properties

(A) The PCE for the variation of exciton binding energy (0–0.45 eV) and diffusion length (0.015–6 μm). (B) The PCE for the variation of exciton diffusivity (0.01–15 cm^2/s) for BEs 0.24 and 0.48 eV. (C) The PCE dependence on the exciton radiative lifetime ($\tau_{\text{ex-r}}$) and exciton nonradiative lifetime ($\tau_{\text{ex-nr}}$) from 0.0001 to 10 ns. (D) Logarithmic scale of the PCE dependence on $\tau_{\text{ex-r}}$ from 0.0001 to 10 ns and $\tau_{\text{ex-nr}}$ from 0.0001 to 0.01 ns.

(BE), exciton radiative lifetime ($\tau_{\text{ex-r}}$), exciton nonradiative lifetime ($\tau_{\text{ex-nr}}$), and exciton diffusion length (EDL) in our model for determining the performance limits of semiconducting 2D TMDC-based PVs. The binding energy of excitons, which describes the ease with which an electron and hole can be separated, is important for solar cell performance. In MoS_2 , the exciton binding energy increases from 0.08 eV³⁴ in the bulk to 0.44 eV³⁵ in monolayers due to quantum confinement effects.³⁶ However, the binding energy in monolayer TMDCs can be modified by molecular coverage,³⁷ doping,³⁸ and engineering of the dielectric environment.^{39,40} By simulating the PCE as a function of exciton binding energy and diffusion length (Figure 2A; Table S2), we found that higher PCE was observed at lower binding energies. This is because excitons are more likely to dissociate at the interface with the aid of thermal energy as the binding energy decreases (Figures S1A–S1C).

EDL is another crucial parameter that influences the transportation of excitons in monolayer TMDCs and, therefore, affects solar cell performance. Since excitons are neutral, they are not significantly affected by electric field drift. In OPVs, the diffusion length is on the nanometer scale, making it unlikely for an exciton to reach an electrode.⁴¹ In contrast, the diffusion length of TMDCs is in the micron range, allowing the electrodes to be further apart and reducing the amount of reflected light. The simulated range of 0.015–6 μm encompasses previously measured diffusion lengths in MoS_2 .⁴² Assuming that excitons follow a Gaussian distribution centered around the diffusion length and are excited evenly throughout the MoS_2 , photocurrent is produced even when the diffusion length is less than the distance between electrodes (1 μm). However, some excitons will recombine before being converted into photocurrent. In silicon PVs, the recombination process mostly involves heat release from nonradiative recombination due to the indirect band gap of silicon. Given that our solar cell design features a direct band gap, a heightened quantum yield results in reduced nonradiative recombination and subsequent heat generation. The emitted photon from radiative recombination may either be

radiated out of the solar cell or reabsorbed. We did not consider the effects of reabsorption⁴³ due to our expectation that they would be low.

The correlation between the performance of a device and its length or diffusion length can be attributed to the exciton density

in the dissociation region. When the EDL is longer than the average distance between the positions where excitons are generated and the regions where dissociation into free charge carriers occurs, increasing the device length further can actually lead to a decrease in exciton densities near the dissociation regions. In Figure 2B (parameters in Table S3), the blue curve, which corresponds to a binding energy of 0.24 eV,⁴⁴ exhibits a decrease in the PCE as the diffusivity increases from 0.01 to 15 cm^2/s . For the red curve, which has a relatively high binding energy of 0.48 eV, a significant drop in the PCE is observed as the diffusivity increases from 0.1 to 1 cm^2/s . However, further increases in diffusivity have a limited impact on the device performance. The respective trends in short-circuit current (I_{SC}), open-circuit voltage (V_{OC}), and fill factor (FF) are presented in the supplemental information (Figures S1D–S1F). For binding energies greater than 0.3 eV, the results for PCE, I_{SC} , V_{OC} , and FF were similar to those obtained for the red curve (see Figures S2A–S2D).

To understand the factors influencing exciton diffusion and dissociation, we examined the radiative lifetime and nonradiative lifetime of the excitons based on a binding energy of 0.24 eV and a diffusion length of 1.5 μm . The quantum yield of monolayer MoS_2 can range from less than 1% to nearly 100%, and the effective lifetime can be as long as 10 ns.⁴⁵ We first conducted simulations varying the exciton radiative lifetime from 0.0001 to 10 ns and the exciton nonradiative lifetime from 0.0001 to 10 ns, which modified the quantum yield from less than 1% to greater than 99% (Figures 2C and S3A–S3C). Previous reports have also found radiative lifetimes as 0.2 ps,⁴⁶ 600 ps,⁴⁷ 1 ns,⁴⁸ and 15 ns⁴⁹ and nonradiative lifetimes from 1 to 10 ps⁴⁸ in MoS_2 . To more closely examine the effect of these lifetimes on PCE, we studied the range with logarithmic axes in Figures 2D and S3D–S3F (parameters in Table S4). The simulations reveal that the total lifetime, which consists of both radiative and nonradiative lifetimes, is the determining factor for photocurrent generation and PCE, as reabsorption was not accounted for in the simulations. A longer lifetime means excitons exist longer before recombining and

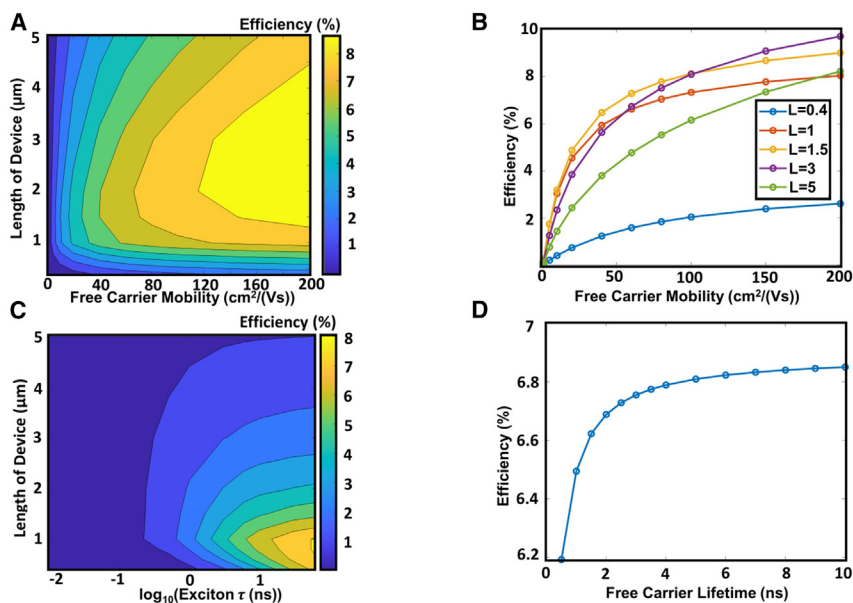


Figure 3. Optimization of physical properties and free carrier transportation

(A) The PCE for the variation of length of device (0.4–5 μm) and free carrier mobility (0.1–200 cm²/(Vs.)) under exciton lifetime of 10 ns.

(B) The PCE for the variation of free carrier mobility for length of device = 0.4, 1, 1.5, 3, and 5 μm.

(C) The PCE for the variation of length of device (0.4–5 μm) and exciton lifetime (0.01–60 ns) under free carrier mobility of 60 cm²/(Vs.).

(D) The PCE for the variation of free carrier lifetime (0.5–10 ns) under free carrier mobility of 60 cm²/(Vs.).

have a higher chance of diffusing and dissociating, leading to a higher photocurrent from an increase in charges. In Figure 2D, improving the nonradiative lifetime is more important than improving the radiative lifetime because defects significantly trap the excitons and reduce their total lifetime, therefore decreasing the efficiency of the PV device. In the optimization process, we selected 10 ns for both radiative and nonradiative lifetimes, resulting in a PCE of 5.62%.

Role of free carriers and their mobilities

After optimizing the parameters of exciton binding energy and exciton lifetimes, we examined the effects of free carrier properties under an electric field on photocurrent. The electron mobility of monolayer MoS₂ has been reported to be in the range of 0.1–10 cm²V^{−1}s^{−1}.^{50,51} However, the carrier mobility can be improved using the dielectric screening effect,⁵² allowing for values greater than 100 cm²V^{−1}s^{−1}.^{53,54} We therefore studied the range of free carrier mobilities from 0.1 to 200 cm²V^{−1}s^{−1} and device lengths from 0.4 to 5 μm (Figures 3A and S4; parameters in Table S5). As the electron mobility increases, the PCE is enhanced as more photocarriers are collected by the electrodes. In Figure 3B, we see that the PCE increases significantly in the mobility range of 0.1–60 cm²V^{−1}s^{−1} for several selected device lengths (0.4, 1, 1.5, 3, and 5 μm). The largest enhancement in PCE is seen when the free carrier mobility is increased to 100 cm²V^{−1}s^{−1} and the radiative lifetime to 6 ns, values that can be achieved in MoS₂ without additional treatments or dielectric engineering.

To understand the impact of electron mobility, we also analyze the effect of device length. In Figure 3A, we see that the peak PCE occurs at a device length around 1.5 μm under the mobility of 60 cm²V^{−1}s^{−1}. Lengthening the device results in a positive correlation with I_{SC}, while exhibiting a negative correlation with FF, as shown in Figures S4A–S4C. For Figures 3C and S4D–S4F, where the exciton total lifetime is varied from 0.01 to 60 ns, the optimized device length for a 60 ns lifetime is compa-

table to the EDL when it is held constant at 1.5 μm during simulations (parameters in Table S6). When the device length is greater than 3 μm, the limited transportation of free charge carriers results in a decrease in PCE. Conversely, when the device length is shorter than 1 μm, the average EDL of 1.5 μm exceeds the disso-

ciation region, leading to a decline in PCE. In Figure 3C, we see that increasing the exciton total lifetime significantly increases the photocurrent and PCE. In addition to carrier mobility, we also studied the carrier lifetime over the range of 0.5–10 ns, where the dominant recombination process at low carrier concentrations is Shockley-Read-Hall (SRH) recombination.⁴⁷ In Figures 3D and S5A–S5C (parameters in Table S7), the PCE increases nonlinearly with carrier lifetime. For SRH lifetimes shorter than 3 ns, the significant increase in PCE shows that the carrier lifetime is a limiting factor for final efficiency. However, for SRH lifetimes longer than 3 ns, the slow increase in PCE indicates that the SRH lifetime is no longer a limiting factor.

Angle sensitivity and comparison between different TMDCs

Photocarrier generation of the superlattice for N = 1 to 10 with optimized bottom alumina thicknesses is investigated at different incident angles. The photocarrier generation rate is found to have a maximum in the range of 50°–65° for all values of N. This increased rate is due to the formation of exciton polaritons, which allows for near-unity absorption of transverse electric polarized light near the photon density maximum of the AM1.5 solar spectrum. However, the absorption at other wavelengths decreases as the incident angle increases due to the initial interface becoming more reflective. The maximum occurs at lower incident angles as N increases because the increased Rabi splitting allows for the formation of exciton polaritons at lower incident angles. Simulations of the absorbed photons at different incident angles were used to calculate the PCE as shown in Figure 4B. For combined contacts, the peak PCE of 0.24 eV binding energy was 9.22% at an incident angle of 55°. The internal quantum efficiency did not change when the incident angle was varied in the simulation using the angle-dependent photon absorption spectra, which explained the same curvature seen in Figures 4A (N = 5) and 4B. The final PCE is proportional to absorbed photons for a constant internal quantum efficiency (IQE).

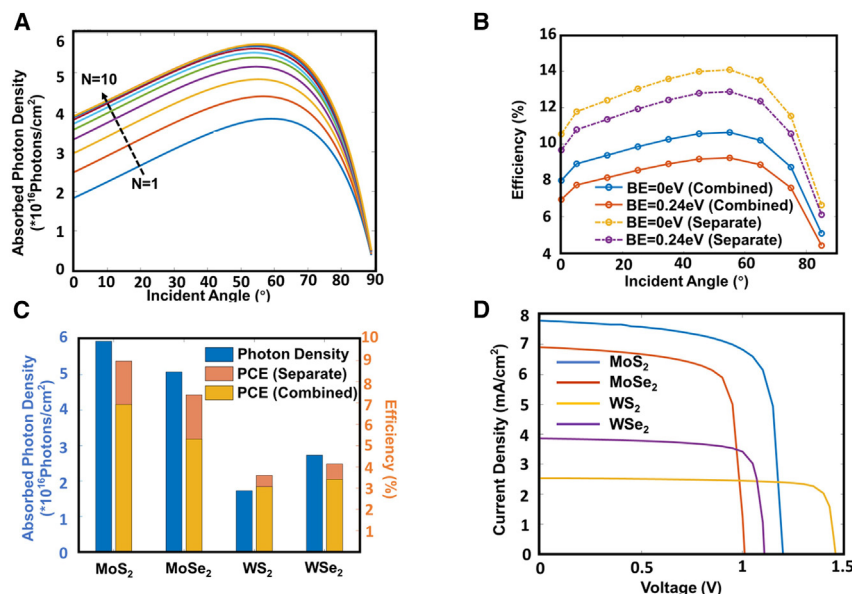


Figure 4. Optimization of incident angles, TMDC materials, and designs

(A) The absorbed photon density for the different incident angle (0–90°) and different layers (N = 1–10). (B) The PCE for the variation of the incident angle (0–90°) for BEs 0 and 0.24 eV with combined contacts (solid lines) and separate contacts (dashed lines).

(C) The absorbed photon density and PCE for MoS₂, MoSe₂, WS₂, and WSe₂ with combined contacts and separate contacts.

(D) The I–V curve for MoS₂, MoSe₂, WS₂, and WSe₂.

Our analysis revealed that the effective performance of the combined contacts in our 5-layer superlattice structure was limited by the lowest performing layer, which was the bottom layer, due to variations in the absorptance of each layer. Therefore, despite increasing the complexity of the structure, we investigated the implementation of separate contacts and found that it leads to an increase in solar cell performance, as depicted in Figures 4B, S10, and S11, and Table S12. Our simulation results indicate that the top layer, which had the highest absorptance and thus the highest I_{SC} and PCE, played a crucial role in achieving high performance in the separate contact configuration. The photocarrier generation rate of the different layers, as presented in Figure S11, supports this observation. Specifically, we found that the highest PCE of 12.87% was attained by the separate contacts when the binding energy was 0.24 eV and the incident angle was 55°. It is important to note, however, that a discrepancy still persists between the efficiency values predicted by our optimized simulation model and the theoretical maximum result detailed balance model. This mainly results from the perfect spectral absorption in the detailed balance model, which does not account for the

spectral absorption profile. The absorbed photon density in monolayer MoS₂ for perfect absorption can be expressed at a rate of 1.38×10^{17} photons per cm², which exceeds the absorbed photon density of our optimized simulation model by a factor of approximately 1.75.

The performance of superlattices made from four different TMDCs, MoS₂, MoSe₂, WS₂, and WSe₂, was also simulated using the combined contact configuration. The absorbed photon density was calculated using TMM simulations, as shown in Table 1 and Figure S13. Our analysis revealed that reducing the exciton binding energy is a critical factor in achieving high solar cell efficiency using this geometry. For the simulations, we assumed a uniform exciton binding energy of 0.24 eV across all materials. The absorbed photon density and PCEs of the different materials are depicted in Figure 4C (parameters in Table S8). Our findings indicate that MoS₂ exhibits the highest PCE and photon absorption density in our superlattice structures due to its two lowest energy excitons being near the maximum of the solar photon spectrum. MoSe₂ also demonstrates a high absorbed photon density owing to its favorable band gap. On the other hand, WS₂ demonstrates a relatively high free carrier mobility of $1,060 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and larger band gap resulting in the highest V_{OC} ; however, its relatively lower photon absorption rate led to a final PCE of 3.07%. Similarly, WSe₂ had a relatively low photon absorption rate, resulting in a PCE of 3.42%. The actual binding energy of excitons in the superlattice is expected to be lower by a factor of approximately 2 since we use the binding energy values for freestanding samples in this analysis. The binding energy can be further reduced by using high-static dielectric co-efficient (k) spacer layers such as TiO₂ and Ta₂O₅. Figure 4D shows the current-voltage (I–V) curves of the four materials. In this comparison, MoS₂ shows the best I_{SC} density of the four TMDCs studied, while WS₂ has the largest V_{OC} . We also note that setting the binding energies to literature-derived values for each TMDC introduces further disparities in the PCE performances. A comparison table and I–V characteristics for various TMDC PVs with literature-derived binding energy values are provided in Note S4.

At a PCE of 9.22% and a specific weight of 0.58 g m^{-2} , excluding a supporting substrate, our combined-contacts device has a specific power (power/weight ratio) of 157 W g^{-1} , which is the highest value among TMDC-based cells. By separating the contacts, the performance improves to a PCE of

Table 1. The key parameters of TMDC materials (MoS₂, MoSe₂, WS₂, WSe₂)

	MoS ₂	MoSe ₂	WS ₂	WSe ₂
Absorbed photons ($\times 10^{16}$ photons/cm ²)	5.94	5.08	1.73	2.73
Band gap (eV)	1.8 ²³	1.60 ⁵⁵	2.04 ⁵⁶	1.65 ⁵⁵
Binding energy (eV)	0.24 ⁴⁴	0.24	0.24	0.24
Mobility (cm ² /V/s)	60 ⁵⁷	480 ⁵⁸	1,060 ⁵⁹	250 ⁶⁰
Short circuit current (mA/cm ²)	7.796	6.91	2.54	3.86
Open circuit voltage (V)	1.20	1.01	1.46	1.11
Fill factor (%)	74.09	76.24	82.83	79.61
Power conversion efficiency (%)	6.93	5.31	3.07	3.42

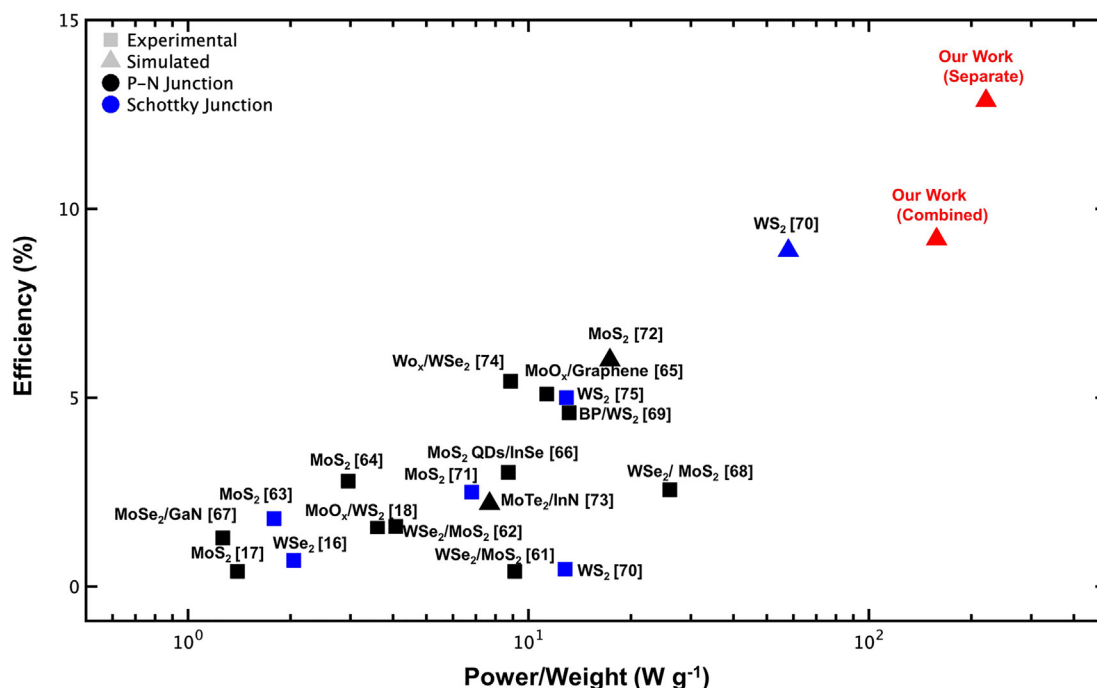


Figure 5. Efficiency comparison of Van der Waals (VDW) material-based PVs with their power/weight ratio in unit of W g^{-1}

12.87% and a specific power of 220 W g^{-1} . We compare the PCE performance and power/weight ratio of our modeled device with other TMDC-containing PV devices from recent literature under AM1.5 illumination (Figure 5).^{16–18,61–75} Further details on the calculation of specific power for each cell are provided in the methods. We found that our solar cell has the largest specific power of all TMDC-based solar cells to the best of our knowledge. This is because previous TMDC-based solar cells have either used monolayer samples for their ideal electronic properties at the expense of lower absorption, or thin films, typically $>100 \text{ nm}$, which have higher absorption but increase the weight. Our design, however, achieves large absorption while also taking advantage of the lightweight, high electrical quality of monolayer TMDCs.

We also compared our solar cell with the highest specific power that has been achieved in other leading technologies^{76–80} (Figure 6). Although our combined-contacts solar cell has the lowest PCE of all the materials at 9.22% where the rest of the materials range from 11.2% to 22.35%, our exceptionally small active layer thickness of 3.5 nm results in TMDCs having the highest potential specific powers of all the materials. Given the simplicity of our design, the ease of fabrication for large-area application, and the excellent power density performance for our TMDC-based solar cell, it is a great candidate for lightweight solar cells in fields such as aerospace and wearable electronics.

Conclusion

In summary, we have proposed a device structure and model for 2D TMDC-based excitonic solar cells and provided a thorough investigation of the physical factors limiting their performance. The principal innovation of our work lies in optimizing both the

optical and electronic properties and accounting for excitonic effects to estimate the highest possible PCE values using practical materials and device parameters. Our findings suggest that the large exciton binding energies limit the overall efficiencies of 2D TMDC-based PV devices to about a third of those predicted by the detailed balance model. Nonetheless, even with large exciton binding energies, upon optimizing various geometric and material quality parameters, PCEs of 9.22% with combined contacts and 12.87% with separate contacts were achieved for an MoS_2 -based device. Combined with a total active layer thickness that is $<4 \text{ nm}$, these PCE values make our proposed devices one of the highest specific power cells ($>100 \text{ W g}^{-1}$) of any thin-film PV technology available today. Overall, our work lays a solid theoretical foundation and computational model for the performance limits of practical and scalable 2D TMDC-based excitonic solar cells.

EXPERIMENTAL PROCEDURES

Resource availability

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Deep Jariwala (dmj@seas.upenn.edu).

Materials availability

This study did not generate new materials.

Data and code availability

The data presented in this paper is available from the corresponding author upon reasonable request.

Methods

Photocurrent generation calculations

The TMM⁸¹ was implemented in python to simulate the photocurrent generation rate of the superlattice due to its ability to accurately model the absorption

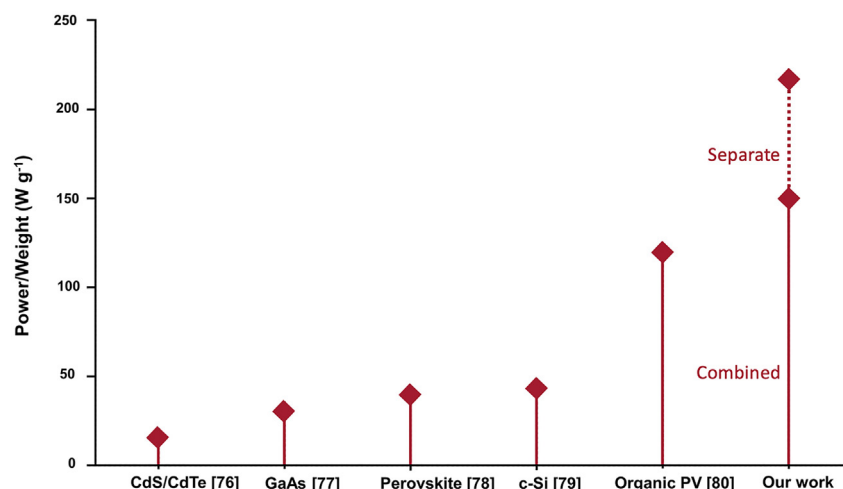


Figure 6. Power/weight ratio chart of different materials in unit of W g⁻¹

spectra of 1D systems. The refractive index of monolayer TMDCs, Al₂O₃, and Au were all taken from literature.^{82,83} The photocarrier generation rate was then calculated by multiplying the absorption spectrum of the superlattice by the AM1.5 solar spectrum.⁸⁴ For nonnormal incidence, the absorption spectra for transverse electric (TE) and transverse magnetic (TM) light were averaged to calculate the absorption spectrum of unpolarized light.

Sentaurus simulation

The 2D solid-state p-i-n superlattice structure was numerically simulated using Sentaurus technology computer-aided design tools from Synopsys. This tool is beneficial in understanding solar cell behavior considering the many physical mechanisms in the device (such as different recombination, different photon absorption calculation). Due to the limitation of vertices number and the long simulation time of the 3D model, a 2D simulation was adopted after we converged on parameters from the 3D model, as shown in Notes S1 and S3, Figures S6–S9, and Tables S9 and S10. The 2D simulation has a default thickness of 1 μm in the third dimension. The thickness variation in the third dimension was not expected to significantly affect the results, and more details can be found in Note S3. The structure consists of the monolayer active materials (MoS₂, MoSe₂, WS₂, WSe₂), insulators (Al₂O₃), cathode (Ag), and anode (Au). The thick bottom insulator and gold bottom were not included in the Sentaurus simulation as their effects were fully considered in the photon generation simulation. The investigation of the impact of binding energy, EDL, exciton radiative lifetime, exciton non-radiative lifetime, free carrier mobility, free carrier SRH lifetime, and device length on V_{OC}, J_{SC}, FF, and PCE have been performed by utilizing this model.

The initial input parameters were obtained from the experimental data and other theoretical results to define the structure and materials as shown in Tables S1–S8. In this model, the electron and hole densities were computed from the electron and hole quasi-Fermi potentials. The band gap was based on reported values, and the band-gap narrowing effect was not considered. This model allowed discontinuous interfaces for a superlattice structure. By defining the heterointerface, the datasets of the two materials were treated properly by introducing double points. The optical generation rate was based on the TMM results and was defined manually for each layer. The carrier recombination considered three forms: SRH, Auger, and radiative. Sentaurus solved the Poisson and continuity equations to account for optical properties. The model coupled the Poisson equation, electron and hole continuity equations, and the singlet exciton equation. The equation modeled the dynamic of the generation, diffusion, recombination, and radiative decay of singlet excitons in the bulk of the device^{85,86} is given by

$$\frac{\partial n_{se}}{\partial t} = R_{bimolec} + \nabla \cdot D_{se} \nabla n_{se} - \frac{n_{se} - n_{se}^{eq}}{\tau} - \frac{n_{se} - n_{se}^{eq}}{\tau_{trap}} - R_{se}, \quad (\text{Equation 1})$$

where n_{se} and n_{se}^{eq} are the singlet exciton and equilibrium exciton densities, respectively; $R_{bimolec}$ is the carrier bimolecular recombination rate acting as

a singlet exciton generation term; D_{se} is the singlet exciton diffusion constant; and τ and τ_{trap} are the singlet exciton lifetimes. R_{se} is the net singlet exciton recombination rate.

The dissociation of excitons has been utilized for the interfaces of P region/i-region and i-region/n-region. Moreover, in order to enhance the performance of this approach, we have explored the effects of incorporating additional dissociation regions. Our results demonstrate that the PCE can be further improved by 0.3% through the incorporation of 11 dissociation interfaces with a uniform distribution. However, it is expected that even a full, detailed analysis of expanding the exciton dissociation regions will not result in significant

enhancement of the PCE. Therefore, this will be explored as part of our future work.

Additional detailed equations are in Note S2 and Table S11.

Specific power estimations

In determining the power/weight ratio for the devices compared in Figures 5 and 6, the specific power (W m⁻²) and specific weight (g m⁻²) are individually calculated. All devices evaluated presented PCEs with the AM1.5 spectrum (integrated power = 1,000 W m⁻²), thus yielding a specific power that is the PCE fraction of the integrated spectrum power. As there lacks a unified way of calculating specific weight in TMDC PVs, this work views the minimum specific weight sufficient to achieve the given PCE as the comparing metric, positioning this approach specifically as a method for comparing thin-film solar cells from a practical optimization perspective. Every layer of a device that actively contributes to power conversion is included in the specific weight, including electrodes, which are included based on areal coverage. For layers that do not directly contribute to power conversion but play a role in the optics of the device, such as dielectrics, metallic reflectors, and oxide-coated substrates, an effective thickness is used, when smaller than the actual layer thickness, equal to light's penetration depth into the material at the peak of the AM1.5 spectrum (500 nm). Substrates that do not play a role in either the electronic or optical behavior of the device, such as polymer films added for device flexibility, are not included in specific weight calculations given that thicknesses vary across devices without uniform logic. The specific weight is found by multiplying the effective thickness of each included layer in the device by the material density, which was found in literature. Note that the full substrate thickness is used in specific device configurations, such as p-n junction cells where the substrate serves as the p/n side. Dividing specific power by specific weight thus yields the power/weight ratio (W g⁻¹).

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.device.2023.100003>.

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AUTHOR CONTRIBUTIONS

Conceptualization, D.J. and J.L.; methodology, J.L., Z.H., L.D., and K.X.; investigation, Z.H., J.L., and L.D.; writing – original draft, Z.H. and L.D.; writing – review & editing, D.J., J.L., Z.H., and L.D.; funding acquisition, D.J.; supervision, D.J.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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REFERENCES

- Efaz, E.T., Rhaman, M.M., Imam, S.A., Bashar, K.L., Kabir, F., Mourtaza, M.E., Sakib, S.N., and Mozahid, F.A. (2021). A review of primary technologies of thin-film solar cells. *Eng. Res. Express* 3, 032001. <https://doi.org/10.1088/2631-8695/ac2353>.
- Reese, M.O., Glynn, S., Kempe, M.D., McGott, D.L., Dabney, M.S., Barnes, T.M., Booth, S., Feldman, D., and Haegel, N.M. (2018). Increasing markets and decreasing package weight for high-specific-power photovoltaics. *Nat. Energy* 3, 1002–1012. <https://doi.org/10.1038/s41560-018-0258-1>.
- Mathews, I., Kantareddy, S.N., Buonassisi, T., and Peters, I.M. (2019). Technology and market perspective for indoor photovoltaic cells. *Joule* 3, 1415–1426. <https://doi.org/10.1016/j.joule.2019.03.026>.
- Seksaria, H., Kaur, A., Singh, K., and De Sarkar, A. (2023). Hexagonal and tetragonal ScX (X = P, As, Sb) nanosheets for optoelectronics and straintronics. *Appl. Surf. Sci.* 675, 156306. <https://doi.org/10.1016/j.apsusc.2022.156306>.
- Sampaio, P.G.V., and González, M.O.A. (2022). A review on organic photovoltaic cell. *Int. J. Energy Res.* 46, 17813–17828. <https://doi.org/10.1002/er.8456>.
- Li, Y., Wang, L., Qiao, Y., Gan, Y., and Singh, D.J. (2020). Prediction of ternary alkaline-earth metal Sn(II) and Pb(II) chalcogenide semiconductors. *Phys. Rev. Mater.* 4, 055004. <https://doi.org/10.1103/PhysRevMaterials.4.055004>.
- Rawat, A., Ahammed, R., Dimple, Jena, N., Mohanta, M.K., and De Sarkar, A. (2019). Solar energy harvesting in type II van der Waals heterostructures of semiconducting group III monochalcogenide monolayers. *J. Phys. Chem. C* 123, 12666–12675. <https://doi.org/10.1021/acs.jpcc.9b03359>.
- Mohanta, M.K., Kishore, A., and De Sarkar, A. (2020). Two-dimensional ultrathin van der Waals heterostructures of indium selenide and boron monophosphide for superfast nanoelectronics, excitonic solar cells, and digital data storage devices. *Nanotechnology* 31, 495208. <https://doi.org/10.1088/1361-6528/abaf20>.
- Liu, L., Xie, Y., Tse, J.S., and Ma, Y. (2023). Theoretical design of two-dimensional AMInP2X3Y3 (AM = Li, Na, K; X/Y = S, Se, Te) monolayers for highly efficient excitonic solar cells. *Mater. Adv.* 4, 570–577. <https://doi.org/10.1039/D2MA00937D>.
- Hou, Y., Wu, C., Yang, D., Ye, T., Honavar, V.G., van Duin, A.C.T., Wang, K., and Priya, S. (2020). Two-dimensional hybrid organic-inorganic perovskites as emergent ferroelectric materials. *J. Appl. Phys.* 128, 060906. <https://doi.org/10.1063/5.0016010>.
- Kang, K., Lee, K.-H., Han, Y., Gao, H., Xie, S., Muller, D.A., and Park, J. (2017). Layer-by-layer assembly of two-dimensional materials into wafer-scale heterostructures. *Nature* 550, 229–233. <https://doi.org/10.1038/nature23905>.
- Laturia, A., Van de Put, M.L., and Vandenbergh, W.G. (2018). Dielectric properties of hexagonal boron nitride and transition metal dichalcogenides: from monolayer to bulk. *npj 2D Mater. Appl.* 2, 6. <https://doi.org/10.1038/s41699-018-0050-x>.
- Amani, M., Lien, D.-H., Kiriya, D., Xiao, J., Azcatl, A., Noh, J., Madhupraty, S.R., Addou, R., Kc, S., Dubey, M., et al. (2015). Near-unity photoluminescence quantum yield in MoS₂. *Science* 350, 1065–1068. <https://doi.org/10.1126/science.122114>.
- Wang, L., Huang, L., Tan, W.C., Feng, X., Chen, L., Huang, X., and Ang, K.-W. (2018). 2D photovoltaic devices: progress and prospects. *Small Methods* 2, 1700294. <https://doi.org/10.1002/smt.201700294>.
- Jariwala, D., Davoyan, A.R., Wong, J., and Atwater, H.A. (2017). Van der Waals materials for atomically-thin photovoltaics: promise and outlook. *ACS Photonics* 4, 2962–2970. <https://doi.org/10.1021/acsp Photonics.7b01103>.
- Akama, T., Okita, W., Nagai, R., Li, C., Kaneko, T., and Kato, T. (2017). Schottky solar cell using few-layered transition metal dichalcogenides toward large-scale fabrication of semitransparent and flexible power generator. *Sci. Rep.* 7, 11967. <https://doi.org/10.1038/s41598-017-12287-6>.
- Li, H.M., Lee, D., Qu, D., Liu, X., Ryu, J., Seabaugh, A., and Yoo, W.J. (2015). Ultimate thin vertical p-n junction composed of two-dimensional layered molybdenum disulfide. *Nat. Commun.* 6, 6564. <https://doi.org/10.1038/ncomms7564>.
- Nassiri Nazif, K., Kumar, A., Hong, J., Lee, N., Islam, R., McClellan, C.J., Kami, O., van de Groep, J., Heinz, T.F., Pop, E., et al. (2021). High-performance p-n junction transition metal dichalcogenide photovoltaic cells enabled by MoO_x doping and passivation. *Nano Lett.* 21, 3443–3450. <https://doi.org/10.1021/acs.nanolett.1c00015>.
- Lin, S., Li, X., Wang, P., Xu, Z., Zhang, S., Zhong, H., Wu, Z., Xu, W., and Chen, H. (2015). Interface designed MoS₂/GaAs heterostructure solar cell with sandwich stacked hexagonal boron nitride. *Sci. Rep.* 5, 15103. <https://doi.org/10.1038/srep15103>.
- Hussain, S., Shaikh, S.F., Vikraman, D., Mane, R.S., Joo, O.-S., Naushad, M., and Jung, J. (2015). Sputtering and sulfurization-combined synthesis of a transparent WS₂ counter electrode and its application to dye-sensitized solar cells. *RSC Adv.* 5, 103567–103572. <https://doi.org/10.1039/C5RA18613G>.
- Kumar, P., Lynch, J., Song, B., Ling, H., Barrera, F., Kisslinger, K., Zhang, H., Anantharaman, S.B., Digani, J., Zhu, H., et al. (2022). Light-matter coupling in large-area van der Waals superlattices. *Nat. Nanotechnol.* 17, 182–189. <https://doi.org/10.1038/s41565-021-01023-x>.
- Palummo, M., Berrios, E., Varsano, D., and Giorgi, G. (2020). Optical properties of lead-free double perovskites by ab initio excited-state methods. *ACS Energy Lett.* 5, 457–463. <https://doi.org/10.1021/acsenenergylett.9b02593>.
- Castellanos-Gomez, A., Barkelid, M., Goossens, A.M., Calado, V.E., van der Zant, H.S.J., and Steele, G.A. (2012). Laser-thinning of MoS₂: on demand generation of a single-layer semiconductor. *Nano Lett.* 12, 3187–3192. <https://doi.org/10.1021/nl301164v>.
- Cai, L., McClellan, C.J., Koh, A.L., Li, H., Yalon, E., Pop, E., and Zheng, X. (2017). Rapid flame synthesis of atomically thin MoO₃ down to monolayer thickness for effective hole doping of WSe₂. *Nano Lett.* 17, 3854–3861. <https://doi.org/10.1021/acs.nanolett.7b01322>.
- Kirchartz, T., Mattheis, J., and Rau, U. (2008). Detailed balance theory of excitonic and bulk heterojunction solar cells. *Phys. Rev. B* 78, 235320. <https://doi.org/10.1103/PhysRevB.78.235320>.
- Roy, S., and Bermel, P. (2022). Tungsten-disulfide-based ultrathin solar cells for space applications. *IEEE J. Photovolt.* 12, 1184–1191. <https://doi.org/10.1109/JPHOTOV.2022.3179986>.
- Roy, S., Hu, Z., Kais, S., and Bermel, P. (2021). Enhancement of photovoltaic current through dark states in donor-acceptor pairs of tungsten-based transition metal di-chalcogenides. *Adv. Funct. Mater.* 31, 2100387. <https://doi.org/10.1002/adfm.202100387>.

28. Khan, M.A.U., Adesina, N.O., and Xu, J. (2022). Near unity absorbance and photovoltaic properties of TMDC/gold heterojunction for solar cell application. *Key Eng. Mater.* 918, 97–105. <https://doi.org/10.4028/p-uz62m4>.
29. Classen, A., Chochos, C.L., Lür, L., Gregoriou, V.G., Wortmann, J., Os-
vet, A., Forberich, K., McCulloch, I., Heumüller, T., and Brabec, C.J. (2020). The role of exciton lifetime for charge generation in organic solar cells at negligible energy-level offsets. *Nat. Energy* 5, 711–719. <https://doi.org/10.1038/s41560-020-00684-7>.
30. Kirchartz, T., and Rau, U. (2008). Charge separation in excitonic and bipolar solar cells — a detailed balance approach. *Thin Solid Films* 516, 7144–7148. <https://doi.org/10.1016/j.tsf.2007.12.084>.
31. Bisquert, J., and Garcia-Belmonte, G. (2011). On voltage, photovoltage, and photocurrent in bulk heterojunction organic solar cells. *J. Phys. Chem. Lett.* 2, 1950–1964. <https://doi.org/10.1021/jz2004864>.
32. Bisquert, J. (2017). *The Physics of Solar Cells: Perovskites, Organics, and Photovoltaic Fundamentals* (CRC Press).
33. Savoie, B.M., Rao, A., Bakulin, A.A., Gelinas, S., Movaghar, B., Friend, R.H., Marks, T.J., and Ratner, M.A. (2014). Unequal partnership: asymmetric roles of polymeric donor and fullerene acceptor in generating free charge. *J. Am. Chem. Soc.* 136, 2876–2884. <https://doi.org/10.1021/ja411859m>.
34. Saigal, N., Sugunakar, V., and Ghosh, S. (2016). Exciton binding energy in bulk MoS₂: a reassessment. *Appl. Phys. Lett.* 108, 132105. <https://doi.org/10.1063/1.4945047>.
35. Hill, H.M., Rigosi, A.F., Roquelet, C., Chernikov, A., Berkelbach, T.C., Reichman, D.R., Hybertsen, M.S., Brus, L.E., and Heinz, T.F. (2015). Observation of excitonic Rydberg states in monolayer MoS₂ and WS₂ by photoluminescence excitation spectroscopy. *Nano Lett.* 15, 2992–2997. <https://doi.org/10.1021/nl504868p>.
36. Bryant, G.W. (1988). Excitons in quantum boxes: correlation effects and quantum confinement. *Phys. Rev. B* 37, 8763–8772. <https://doi.org/10.1103/PhysRevB.37.8763>.
37. Wang, K., and Paulus, B. (2020). Tuning the binding energy of excitons in the MoS₂ monolayer by molecular functionalization and defective engineering. *Phys. Chem. Chem. Phys.* 22, 11936–11942. <https://doi.org/10.1039/d0cp01239d>.
38. Li, Z., Xiao, Y., Gong, Y., Wang, Z., Kang, Y., Zu, S., Ajayan, P.M., Nord-
lander, P., and Fang, Z. (2015). Active light control of the MoS₂ monolayer exciton binding energy. *ACS Nano* 9, 10158–10164. <https://doi.org/10.1021/acsnano.5b03764>.
39. Raja, A., Chaves, A., Yu, J., Arefe, G., Hill, H.M., Rigosi, A.F., Berkelbach, T.C., Nagler, P., Schüller, C., Korn, T., et al. (2017). Coulomb engineering of the bandgap and excitons in two-dimensional materials. *Nat. Commun.* 8, 15251. <https://doi.org/10.1038/ncomms15251>.
40. Kishore, A., Tripathy, N., and De Sarkar, A. (2023). Unconventional anisotropy in excitonic properties and carrier mobility in iodine-based XTe (X = Ga, In) monolayers for visible-light photocatalytic water splitting. *J. Phys. Chem. C* 127, 1992–2002. <https://doi.org/10.1021/acs.jpcc.2c07367>.
41. Luhman, W.A., and Holmes, R.J. (2011). Investigation of energy transfer in organic photovoltaic cells and impact on exciton diffusion length measurements. *Adv. Funct. Mater.* 21, 764–771. <https://doi.org/10.1002/adfm.201001928>.
42. Uddin, S.Z., Kim, H., Lorenzon, M., Yeh, M., Lien, D.H., Barnard, E.S., Htoon, H., Weber-Bargioni, A., and Javey, A. (2020). Neutral exciton diffusion in monolayer MoS₂. *ACS Nano* 14, 13433–13440. <https://doi.org/10.1021/acsnano.0c05305>.
43. Yamada, T., Yamada, Y., Nakaike, Y., Wakamiya, A., and Kanemitsu, Y. (2017). Photon emission and reabsorption processes in CH₃NH₃PbBr₃ single crystals revealed by time-resolved two-photon-excitation photoluminescence microscopy. *Phys. Rev. Appl.* 7, 014001. <https://doi.org/10.1103/PhysRevApplied.7.014001>.
44. Park, S., Mutz, N., Schultz, T., Blumstengel, S., Han, A., Aljarb, A., Li, L.-J., List-Kratochvil, E.J.W., Amsalem, P., and Koch, N. (2018). Direct determi-
nation of monolayer MoS₂ and WSe₂ exciton binding energies on insulating and metallic substrates. *2D Mater.* 5, 025003. <https://doi.org/10.1088/2053-1583/aaa4ca>.
45. Lien, D.-H., Uddin, S.Z., Yeh, M., Amani, M., Kim, H., Ager, J.W., Yablono-
vitch, E., and Javey, A. (2019). Electrical suppression of all nonradiative recombination pathways in monolayer semiconductors. *Science* 364, 468–471. <https://doi.org/10.1126/science.aaw8053>.
46. Wang, H., Zhang, C., Chan, W., Manolatos, C., Tiwari, S., and Rana, F. (2016). Radiative lifetimes of excitons and trions in monolayers of the metal dichalcogenide MoS₂. *Phys. Rev. B* 93, 045407. <https://doi.org/10.1103/PhysRevB.93.045407>.
47. Salehzadeh, O., Tran, N.H., Liu, X., Shih, I., and Mi, Z. (2014). Exciton kinetics, quantum efficiency, and efficiency droop of monolayer MoS₂ light-emitting devices. *Nano Lett.* 14, 4125–4130. <https://doi.org/10.1021/nl5017283>.
48. Palummo, M., Bernardi, M., and Grossman, J.C. (2015). Exciton radiative lifetimes in two-dimensional transition metal dichalcogenides. *Nano Lett.* 15, 2794–2800. <https://doi.org/10.1021/nl503799t>.
49. Kim, H., Lien, D.-H., Amani, M., Ager, J.W., and Javey, A. (2017). Highly stable near-unity photoluminescence yield in monolayer MoS₂ by fluoro-polymer encapsulation and superacid treatment. *ACS Nano* 11, 5179–5185. <https://doi.org/10.1021/acsnano.7b02521>.
50. Kim, T., Fan, S., Lee, S., Joo, M.K., and Lee, Y.H. (2020). High-mobility junction field-effect transistor via graphene/MoS₂ heterointerface. *Sci. Rep.* 10, 13101. <https://doi.org/10.1038/s41598-020-70038-6>.
51. Liu, H., Neal, A.T., and Ye, P.D. (2012). Channel length scaling of MoS₂ MOSFETs. *ACS Nano* 6, 8563–8569. <https://doi.org/10.1021/nn303513c>.
52. Rai, A., Valsaraj, A., Movva, H.C.P., Roy, A., Tutuc, E., Register, L.F., and Banerjee, S.K. (2015). Interfacial-Oxygen-Vacancy Mediated Doping of MoS₂ by High-κ Dielectrics. 73rd Annual Device Research Conference, 189–190. <https://doi.org/10.1109/DRC.2015.7175626>.
53. Choi, Y., Kim, H., Yang, J., Shin, S.W., Um, S.H., Lee, S., Kang, M.S., and Cho, J.H. (2018). Proton-conductor-gated MoS₂ transistors with room temperature electron mobility of >100 cm² V⁻¹ s⁻¹. *Chem. Mater.* 30, 4527–4535. <https://doi.org/10.1021/acs.chemmater.8b00568>.
54. Ratan, A., Kunchakara, S., Dutt, M., Tripathi, A., and Singh, V. (2019). 100 MeV Silicon⁹⁺ swift heavy ion irradiation - strategic defect annealing approach to enhance the electrical conductivity of few-layered MoS₂ sheets - PVA nanocomposite film. *Vacuum* 169, 108939. <https://doi.org/10.1016/j.vacuum.2019.108939>.
55. He, K., Kumar, N., Zhao, L., Wang, Z., Mak, K.F., Zhao, H., and Shan, J. (2014). Tightly bound excitons in monolayer WSe₂. *Phys. Rev. Lett.* 113, 026803. <https://doi.org/10.1103/PhysRevLett.113.026803>.
56. Cheng, G., Li, B., Zhao, C., Jin, Z., Li, H., Lau, K.M., and Wang, J. (2019). Exciton aggregation induced photoluminescence enhancement of monolayer WS₂. *Appl. Phys. Lett.* 114, 232101. <https://doi.org/10.1063/1.5096206>.
57. Huo, N., Yang, Y., Wu, Y.N., Zhang, X.G., Pantelides, S.T., and Konstantatos, G. (2018). High carrier mobility in monolayer CVD-grown MoS₂ through phonon suppression. *Nanoscale* 10, 15071–15077. <https://doi.org/10.1039/c8nr04416c>.
58. Kumar, N., Cui, Q., Ceballos, F., He, D., Wang, Y., and Zhao, H. (2014). Exciton diffusion in monolayer and bulk MoSe₂. *Nanoscale* 6, 4915–4919. <https://doi.org/10.1039/C3NR06863C>.
59. Rawat, A., Jena, N., Dimple, D., and De Sarkar, A. (2018). A comprehensive study on carrier mobility and artificial photosynthetic properties in group VI B transition metal dichalcogenide monolayers. *J. Mater. Chem. A Mater.* 6, 8693–8704. <https://doi.org/10.1039/C8TA01943F>.
60. Cui, Q., Ceballos, F., Kumar, N., and Zhao, H. (2014). Transient absorption microscopy of monolayer and bulk WSe₂. *ACS Nano* 8, 2970–2976. <https://doi.org/10.1021/nn500277y>.
61. Wong, J., Jariwala, D., Tagliabue, G., Tat, K., Davoyan, A.R., Sherrott, M.C., and Atwater, H.A. (2017). High photovoltaic quantum efficiency in

- ultrathin van der Waals heterostructures. *ACS Nano* 11, 7230–7240. <https://doi.org/10.1021/acs.nano.7b03148>.
62. Yang, S., Cha, J., Kim, J.C., Lee, D., Huh, W., Kim, Y., Lee, S.W., Park, H.-G., Jeong, H.Y., Hong, S., et al. (2020). Monolithic interface contact engineering to boost optoelectronic performances of 2D semiconductor photovoltaic heterojunctions. *Nano Lett.* 20, 2443–2451. <https://doi.org/10.1021/acs.nanolett.9b05162>.
 63. Shanmugam, M., Durcan, C.A., and Yu, B. (2012). Layered semiconductor molybdenum disulfide nanomembrane based Schottky-barrier solar cells. *Nanoscale* 4, 7399–7405. <https://doi.org/10.1039/C2NR32394J>.
 64. Wi, S., Kim, H., Chen, M., Nam, H., Guo, L.J., Meyhofer, E., and Liang, X. (2014). Enhancement of photovoltaic response in multilayer MoS₂ induced by plasma doping. *ACS Nano* 8, 5270–5281. <https://doi.org/10.1021/nn5013429>.
 65. Nassiri Nazif, K., Daus, A., Hong, J., Lee, N., Vaziri, S., Kumar, A., Nitta, F., Chen, M.E., Kananian, S., Islam, R., et al. (2021). High-specific-power flexible transition metal dichalcogenide solar cells. *Nat. Commun.* 12, 7034. <https://doi.org/10.1038/s41467-021-27195-7>.
 66. Ulaganathan, R.K., Yadav, K., Sankar, R., Chou, F.C., and Chen, Y.-T. (2019). Hybrid InSe nanosheets and MoS₂ quantum dots for high-performance broadband photodetectors and photovoltaic cells. *Adv. Mater. Interfaces* 6, 1801336. <https://doi.org/10.1002/admi.201801336>.
 67. Chen, Z., Liu, H., Chen, X., Chu, G., Chu, S., and Zhang, H. (2016). Wafer-size and single-crystal MoSe₂ atomically thin films grown on GaN substrate for light emission and harvesting. *ACS Appl. Mater. Interfaces* 8, 20267–20273. <https://doi.org/10.1021/acsami.6b04768>.
 68. Tsai, M.-L., Li, M.-Y., Retamal, J.R.D., Lam, K.-T., Lin, Y.-C., Suenaga, K., Chen, L.-J., Liang, G., Li, L.-J., and He, J.-H. (2017). Single atomically sharp lateral monolayer p-n heterojunction solar cells with extraordinarily high power conversion efficiency. *Adv. Mater.* 29, 1701168. <https://doi.org/10.1002/adma.201701168>.
 69. Kwak, D.-H., Ra, H.-S., Jeong, M.-H., Lee, A.-Y., and Lee, J.-S. (2018). High-performance photovoltaic effect with electrically balanced charge carriers in black phosphorus and WS₂ heterojunction. *Adv. Mater. Interfaces* 5, 1800671. <https://doi.org/10.1002/admi.201800671>.
 70. Went, C.M., Wong, J., Jahelka, P.R., Kelzenberg, M., Biswas, S., Hunt, M.S., Carbone, A., and Atwater, H.A. (2019). A new metal transfer process for van der Waals contacts to vertical Schottky-junction transition metal dichalcogenide photovoltaics. *Sci. Adv.* 5, eaax6061. <https://doi.org/10.1126/sciadv.aax6061>.
 71. Fontana, M., Deppe, T., Boyd, A.K., Rinzan, M., Liu, A.Y., Paranjape, M., and Barbara, P. (2013). Electron-hole transport and photovoltaic effect in gated MoS₂ Schottky junctions. *Sci. Rep.* 3, 1634. <https://doi.org/10.1038/srep01634>.
 72. Svatek, S.A., Bueno-Blanco, C., Lin, D.-Y., Kerfoot, J., Macias, C., Zehender, M.H., Tobias, I., García-Linares, P., Taniguchi, T., Watanabe, K., et al. (2021). High open-circuit voltage in transition metal dichalcogenide solar cells. *Nano Energy* 79, 105427. <https://doi.org/10.1016/j.nanoen.2020.105427>.
 73. Villegas, C.E.P., and Rocha, A.R. (2015). Elucidating the optical properties of novel heterolayered materials based on MoTe₂-InN for photovoltaic applications. *J. Phys. Chem. C* 119, 11886–11895. <https://doi.org/10.1021/jp5122596>.
 74. Kim, K.H., Andreev, M., Choi, S., Shim, J., Ahn, H., Lynch, J., Lee, T., Lee, J., Nazif, K.N., Kumar, A., et al. (2022). High-efficiency WSe₂ photovoltaic devices with electron-selective contacts. *ACS Nano* 16, 8827–8836. <https://doi.org/10.1021/acs.nano.1c10054>.
 75. Wang, H., Wang, W., Zhong, Y., Li, D., Li, Z., Xu, X., Song, X., Chen, Y., Huang, P., Mei, A., et al. (2022). Approaching the external quantum efficiency limit in 2D photovoltaic devices. *Adv. Mater.* 34, 2206122. <https://doi.org/10.1002/adma.202206122>.
 76. Gupta, A., Parikh, V., and Compaan, A.D. (2006). High efficiency ultra-thin sputtered CdTe solar cells. *Sol. Energy Mater. Sol. Cells* 90, 2263–2271. <https://doi.org/10.1016/j.solmat.2006.02.029>.
 77. Chen, H.-L., Cattoni, A., De Lépinay, R., Walker, A.W., Höhn, O., Lackner, D., Siefert, G., Faustini, M., Vandamme, N., Goffard, J., et al. (2019). A 19.9%-efficient ultrathin solar cell based on a 205-nm-thick GaAs absorber and a silver nanostructured back mirror. *Nat. Energy* 4, 761–767. <https://doi.org/10.1038/s41560-019-0434-y>.
 78. Kaltenbrunner, M., Adam, G., Giowacki, E.D., Drack, M., Schwödiauer, R., Leonat, L., Apaydin, D.H., Groiss, H., Scharber, M.C., White, M.S., et al. (2015). Flexible high power-per-weight perovskite solar cells with chromium oxide-metal contacts for improved stability in air. *Nat. Mater.* 14, 1032–1039. <https://doi.org/10.1038/nmat4388>.
 79. Xue, M., Nazif, K.N., Lyu, Z., Jiang, J., Lu, C.-Y., Lee, N., Zang, K., Chen, Y., Zheng, T., Kamins, T.I., et al. (2020). Free-standing 2.7 μm thick ultrathin crystalline silicon solar cell with efficiency above 12.0%. *Nano Energy* 70, 104466. <https://doi.org/10.1016/j.nanoen.2020.104466>.
 80. Xiong, S., Fukuda, K., Lee, S., Nakano, K., Dong, X., Yokota, T., Tajima, K., Zhou, Y., and Someya, T. (2022). Ultrathin and efficient organic photovoltaics with enhanced air stability by suppression of zinc element diffusion. *Adv. Sci.* 9, 2105288. <https://doi.org/10.1002/advs.202105288>.
 81. Pettersson, L.A.A., Roman, L.S., and Inganäs, O. (1999). Modeling photo-current action spectra of photovoltaic devices based on organic thin films. *J. Appl. Phys.* 86, 487–496. <https://doi.org/10.1063/1.370757>.
 82. Hsu, C., Frisenda, R., Schmidt, R., Arora, A., Vasconcellos, S.M., Bratschitsch, R., Zant, H.S.J., and Castellanos-Gomez, A. (2019). Thickness-Dependent Refractive Index of 1L, 2L, and 3L MoS₂, MoSe₂, WS₂, and WSe₂. *Adv. Opt. Mater.* 7, 1900239. <https://doi.org/10.1002/adom.201900239>.
 83. McPeak, K.M., Jayanti, S.V., Kress, S.J.P., Meyer, S., Iotti, S., Rossinelli, A., and Norris, D.J. (2015). Plasmonic films can easily be better: rules and recipes. *ACS Photonics* 2, 326–333. <https://doi.org/10.1021/ph5004237>.
 84. National Renewable Energy Laboratory (1962). Reference Air Mass 1.5 Spectra. <https://www.nrel.gov/grid/solar-resource/spectra-am1.5.html>.
 85. Ruhstaller, B., Beierlein, T., Riel, H., Karg, S., Scott, J.C., and Riess, W. (2003). Simulating electronic and optical processes in multilayer organic light-emitting devices. *IEEE J. Sel. Top. Quantum Electron.* 9, 723–731. <https://doi.org/10.1109/JSTQE.2003.818852>.
 86. Ruhstaller, B., Carter, S.A., Barth, S., Riel, H., Riess, W., and Scott, J.C. (2001). Transient and steady-state behavior of space charges in multilayer organic light-emitting diodes. *J. Appl. Phys.* 89, 4575–4586. <https://doi.org/10.1063/1.1352027>.